

HySoP: Hybrid Simulation with Particles

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Summary

The HySoP numerical tool is a Python package designed for high-performance numerical simulations of fluid-related problems based on semi-Lagrangian Remeshed Particle Methods (RPMs). This tool, dedicated to incompressible flows, has been successfully used for external flows, including cases involving bluff bodies, as well as for flows coupled with transport equations and flows through porous media.

In the Computational Fluid Dynamics community, the most widely used methods – such as finite difference, finite volume, finite element and spectral methods – operate within a purely Eulerian framework and have been extensively studied both in terms of consistency and stability, as well as numerical dissipation characterization.

Meanwhile, particle-based approaches have experienced significant development in this context. Among the most popular particle methods, one can cite the Smooth Particle Hydrodynamics (SPH) methods, mostly used to solve the weakly compressible Navier-Stokes equations in velocity-pressure formulation, and the Vortex Methods (VMs), which are generally used to solve the incompressible Navier-Stokes equations in their velocity-vorticity formulation. These methods differ from Eulerian approaches due to their intuitive (meshless) Lagrangian description of the fluid flow, as well as their low numerical dissipation and their ability to circumvent the nonlinearities related to the advection phenomenon. Many efforts have been devoted to addressing the main intrinsic difficulties of purely Lagrangian particle methods. These efforts have led, in particular, to the design of semi-Lagrangian approaches – also referred to as Remeshed Particle Methods (RPMs) – where the particles discretizing the flow are regularly remeshed onto a Cartesian grid. The present HySoP library is based on an RPM to handle this semi-Lagrangian approach through the « particle remeshing » key step on top of the VM framework. This approach combines the strengths of both Lagrangian Vortex methods and Eulerian schemes ([Mimeau & Mortazavi, 2021](#)).

Statement of need

The HySoP library (Hybrid Simulation with Particles) is designed for distributed hybrid architectures, supporting CPU and GPU compute devices through MPI+OpenCL. Its high-level functionalities and the user interface are mainly written in Python. In general, HySoP aims to provide an architecture-agnostic, performance-portable, and easily reusable numerical code. It should be noted that HySoP is scientific research software under continuous development. Among the most advanced open-source libraries related to RPM and similar to HySoP are OpenFPM ([Incardona et al., 2019](#)) and Murphy ([Gillis & Rees, 2022](#)). Both are parallel

and accelerated libraries. OpenFPM is an open-source C++ framework designed for parallel particles-only and hybrid particle-mesh codes. The interpolation schemes it provides are built to support various types of meshes, whereas HySoP relies solely on simple one-dimensional interpolations along the directions of a Cartesian grid. Murphy is a multiresolution adaptive grid framework for numerical simulations on a 3D block-structured collocated grids designed for distributed computational architectures. This multiresolution finds its benefits in dealing with unbounded domains, while HySoP is restricted to a homogeneous Cartesian grid that does not require interpolations or handling computational load imbalance.

Features of the software

Description of the software design

HySoP has been designed on the basis of a decoupling between the mathematical specifications of the problem to solve and the numerical methods and algorithms. The goal is to allow the user to specify only the higher-level specifications, using a formulation that remains close to the mathematical formalism:

- problem parameters;
- computational domain;
- variables defined on the domain;
- operators;
- overall discretisation for the Cartesian grid.

Specifying lower-level details like numerical methods, target architectures, and parallelism layout is optional. The corresponding code is interchangeable and upgradable without requiring any changes to the user's code.

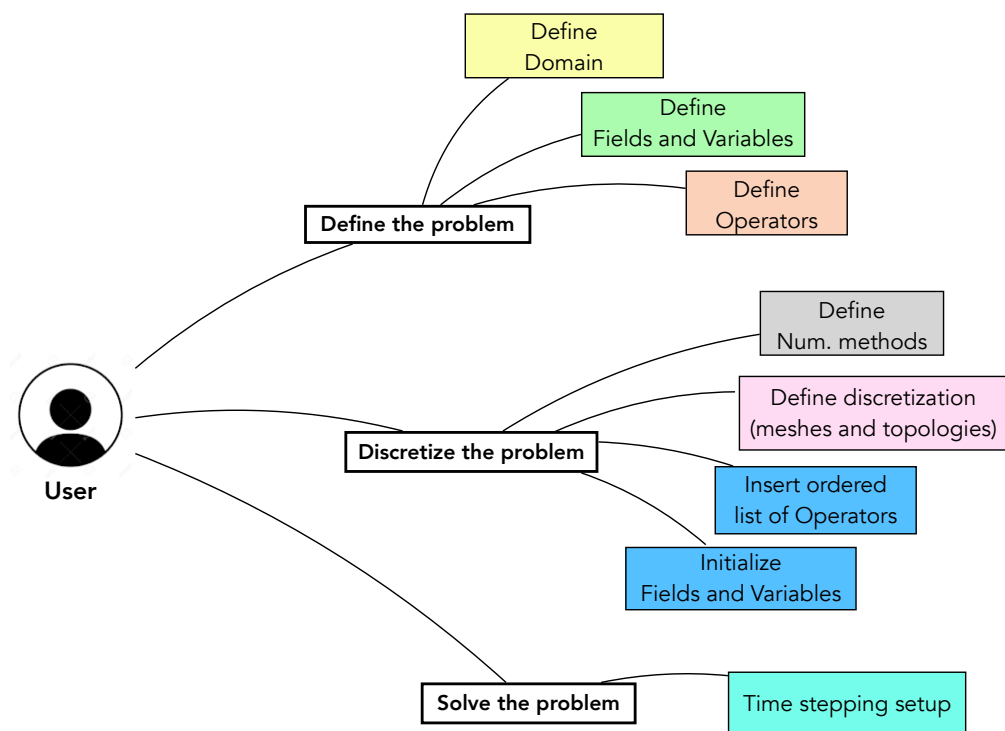


Figure 1: Use case diagram

From the user's perspective, the main use case can be decomposed into three main steps (cf Figure 1):

1. Problem description: expressed as a mathematical PDE formalism using the domain, variables, and operators;
2. Problem initialization: after describing numerical methods and their parameters, the user specifies the main grid resolution, the mesh decomposition for parallel execution, and the compute backend. Then, the user defines the operator ordering and data initialization procedures. At this stage, all memory allocations are completed, the generated code is optimized and compiled, and the communication layout is established.
3. Problem solving: For time-dependent problems, the user provides additional parameters (e.g., time steps) and computations proceed by sequentially applying the operators.

Programming languages and external dependencies

The main drawback of pure Python programs is performance, which is mitigated by using arrays and tools from the numpy module. Additional performance improvements are achieved through optimized external libraries like FFTW for fast Fourier transforms. HySoP is also using the f2py Python module to use a Fortran implementation of RPM ((Lagaert et al., 2014)). Further speedup is obtained via just-in-time compiling using numba or OpenCL towards multi-core architectures. Numba translates Python code into compiled code at runtime, while OpenCL requires explicit code. In HySoP, this code is generated from a formal representation of the numerical methods. Additionally, micro-benchmarks are performed during initialization to optimize the parameters of the generated OpenCL code.

The interaction between the computing backends and the base Python layer is summarized in Figure 2. Our software architecture allows easy integration of new backends, for which ensuring complete interoperability remains the main challenge.

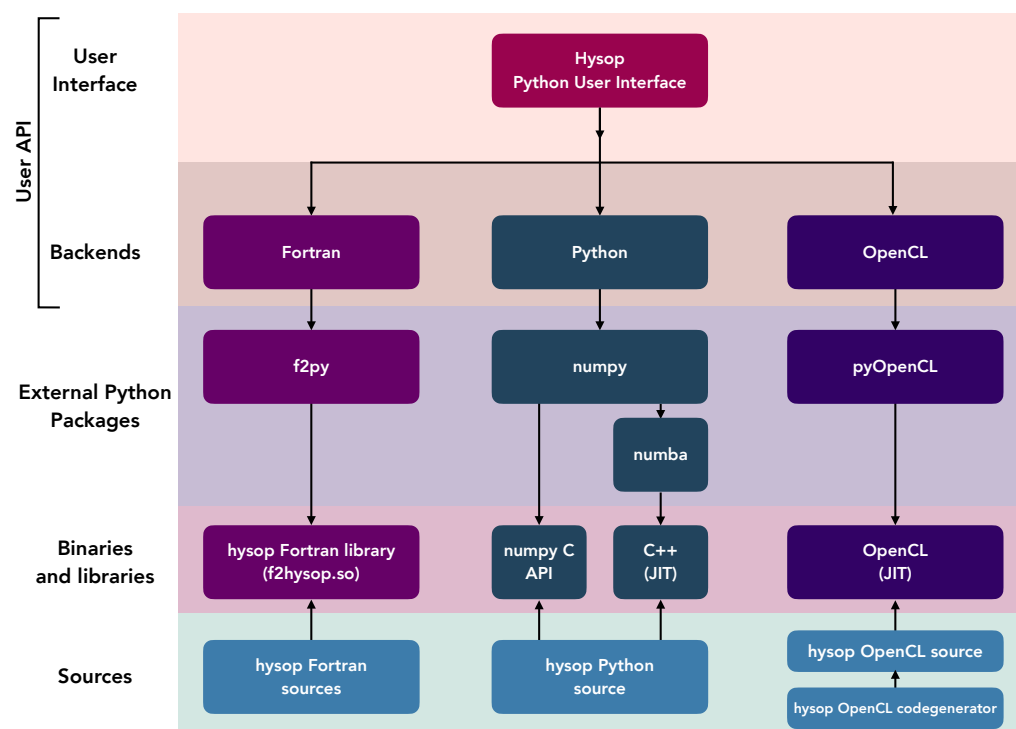


Figure 2: Computing backends in HySoP

Parallelism

The software supports distributed memory parallelism through domain decomposition, relying on MPI parallelism through the mpi4py interface. As a result, HySoP may be considered a Python-based solver that utilizes hybrid MPI+OpenCL programming for heterogeneous computing platforms.

HySoP also exploits parallelism through an operator splitting approach, enabling coarse-grained task parallelism by executing simultaneously distinct and weakly coupled operators, as defined in the overall mathematical formulation.

Evaluating HySoP's computational performance is challenging; however, references such as (Cottet et al., 2014), (Keck, 2019), and (Keck et al., 2021) describe performance insights for multi-GPU and heterogeneous platform simulations.

Applications

The following list illustrates the successful use of the HySoP library across a wide range of domains of applications, each involving diverse governing equations. This illustrates its versatility and flexibility:

Applications	Involved equations	Reference
- Bluff body flows	Navier-Stokes	(Mimeau et al., 2016, 2021)
- Large-Eddy Simulations (sub-grid scale modeling)	Filtered Navier-Stokes	(Crouy-Chanel et al., 2024)
- Transport of passive scalar at high Schmidt number	Navier-Stokes and a passive scalar advection-diffusion	(Cottet et al., 2014)
- Sedimentation in high Schmidt number flows	Navier-Stokes coupled with scalars advection-diffusion	(Keck et al., 2021)
- Passive flow control using porous media	Brinkman-Navier-Stokes	(Mimeau et al., 2017)
- Porous media dissolution at pore-scale	Darcy-Brinkman-Stokes coupled with reactive transport	(Etancelin et al., 2020)
- Porous media precipitation and crystallization at pore-scale	Darcy-Brinkman-Stokes coupled with reactive transport	(Perez et al., 2025)

For Navier-Stokes flows, HySoP relies on the operator splitting to use the most efficient numerical methods for convection terms (RPM) and diffusion terms (Fourier solvers). Scalar transport coupled with fluid flow exploits the spatial scale separation induced by large Schmidt numbers using bi-level discretization on nested Cartesian grids. Porous media are handled by the Brinkman penalization method that enables flows around permeable bodies or internal flows through porous media whose possibly complex structure is difficult to capture using traditional meshes. Finally, in the context of weakly coupled sub-problems, HySoP takes advantage of task parallelism by concurrently using several architectures (e.g., flow solver on CPUs and reactive transport on GPUs).

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